

Readout–scale competition

Why the residual may become harder faster than the geometry improves

Working explanation. Readout fitting can remove coarse error before hidden scales grow appreciably. The remaining residual then has weak independent scale sensitivity, while bounded slopes leave additional accuracy encoded in poorly resolved feature directions. Persistence along joint tanh training is the missing theorem.

1 Residual refinement and Fourier sensitivity

Use the raw network and averaged sample loss

$$f(x) = d + \sum_{j=1}^W c_j \tanh(a_j x + b_j), \quad L = \frac{1}{2m} \sum_i [f(x_i) - g(x_i)]^2. \quad (1)$$

Here c_j are readout weights, $\gamma_j = |a_j|$ inverse widths, and $z_j = -b_j/a_j$ centers. Larger γ_j means narrower transitions. Write $e = f - g$, $r_i = e(x_i)/\sqrt{m}$, and $R = \|r\|_2$. With reference spacing h , $\lambda_j = h\gamma_j$ measures localization relative to that spacing.

The frequency principle identifies weak high-frequency gradients for small tanh slopes [1, Section 6]. Multistage-network experiments document increasingly small, oscillatory residuals and rescale later stages accordingly [2, Section 2.2]. This need not mean high-frequency error grows: in a two-mode model with $\dot{E}_\ell = -\mu_\ell E_\ell$, $\mu_L > \mu_H$, its relative share grows as $|E_H/E_L| \propto e^{(\mu_L - \mu_H)t}$ even at fixed geometry.

For a center-preserving scale derivative, define $\psi_{\gamma,z}(x) = (x-z) \operatorname{sech}^2(\gamma(x-z))$. With $\hat{u}(\omega) = \int u(x) e^{-i\omega x} dx$ and $F(\xi) = \pi\xi / \sinh(\pi\xi/2)$,

$$\partial_\gamma L_{\mathbb{R}} = \frac{c}{2\pi} \int \hat{e}(\omega) \overline{\hat{\psi}_{\gamma,z}(\omega)} d\omega, \quad \hat{\psi}_{\gamma,z}(\omega) = \frac{ie^{-i\omega z}}{\gamma^2} F'(\omega/\gamma). \quad (2)$$

This is for a differentiable whole-line loss $L_{\mathbb{R}} = \frac{1}{2} \int e^2$. If $e \in L^2$, $\hat{e} \in L^1$, and $\hat{e} = 0$ on $|\omega| < \Omega$, with $\Omega \geq \gamma$, then

$$|\partial_\gamma L_{\mathbb{R}}| \leq \frac{C_F |c| M_\Omega \Omega}{\gamma^3} e^{-\pi\Omega/(2\gamma)}, \quad M_\Omega = \frac{1}{2\pi} \int_{|\omega| \geq \Omega} |\hat{e}(\omega)| d\omega. \quad (3)$$

One may take $C_F = \pi^2 \coth(\pi/2)/(1 - e^{-\pi})$, by bounding $|F'(\xi)|$ for $|\xi| \geq 1$. Thus a substantial residual can give exponentially weak scale signal when its frequency is large *relative to current* γ . Low-frequency leakage, density weighting, finite-window tails, and sampling require separate control; (3) is not automatically a bound for (1).

2 Readout compensation and sensitivity squaring

Let A be the normalized sample–feature matrix, including the constant column; $v = (d, c)$ and $y_i = g(x_i)/\sqrt{m}$, so $r = Av - y$. Let $J = \partial_\theta(Av)$ at fixed readout, with $\theta = (a, b)$. Set $P = AA^\dagger$, $C = A^\dagger J$, and $Z = (I - P)J$. Since $J = AC + Z$,

$$\nabla_\theta L = C^T \nabla_v L + Z^T (I - P) r. \quad (4)$$

The first term is geometry force shared with readout fitting. At an exact LS readout it vanishes: for $\Phi(\theta) = \min_v L$, $\nabla \Phi = Z_*^T r_*$, locally at constant rank [3]. Near an exact fit, $r_* \simeq Z_* \Delta\theta$, so $\nabla \Phi \simeq Z_*^T Z_* \Delta\theta$: singular sensitivity s gives a GD rate s^2 . Fourier attenuation and readout compensation can both make sensitivity small; GD multiplies forward and backward sensitivities. No high-frequency assumption is needed for this local identity, nor is $\|\nabla \Phi\|$ universally proportional to R^2 .

3 An exact model of readout winning the race

Let $u(x)$ be a fixed unit-RMS function (for example $\sqrt{3}x$ on uniform $[-1, 1]$). Define the target $g = \beta u$, with amplitude $\beta > 0$, and the trainable prediction $f = c(t)\gamma(t)u$. This is an exact bilinear model of the leading broad-tanh term $c \tanh(\gamma u) \simeq c\gamma u$, not a globally accurate replacement for $\tanh$.

With initial values $c_0, \gamma_0 > 0$, $c_0\gamma_0 < \beta$, and learning-rate factors $\eta_c, \eta_\gamma > 0$, the loss and coupled gradient flow are

$$L = \frac{1}{2}(c\gamma - \beta)^2, \quad \dot{c} = \eta_c\gamma(\beta - c\gamma), \quad \dot{\gamma} = \eta_\gamma c(\beta - c\gamma). \quad (5)$$

Put $\nu = \eta_\gamma/\eta_c$. The conserved quantity is $\gamma^2 - \nu c^2 = D := \gamma_0^2 - \nu c_0^2$. Combining it with $c_\infty\gamma_\infty = \beta$ yields

$$\boxed{\gamma_\infty^2 = \frac{D + \sqrt{D^2 + 4\nu\beta^2}}{2}}. \quad (6)$$

As $\eta_c/\eta_\gamma \rightarrow \infty$, $\gamma_\infty \rightarrow \gamma_0$: the readout fits the coarse component while the scale barely moves. The toy target itself is fitted exactly.

Freezing at $c_f = c(t_f) > 0$ gives $\gamma_\infty = \beta/c_f$, with exponential relaxation rate $\eta_\gamma c_f^2$. Earlier freezing leaves a smaller c_f , so it selects a larger eventual scale but can take longer to reach it.

Bridge to actual tanh. For $\phi_j = (\tanh(\gamma_j(x_i - z_j))/\sqrt{m})_i$, define $M_{3,j} = (m^{-1} \sum_i |x_i - z_j|^6)^{1/2}$. Set $\zeta_j = \partial_{\gamma_j} \phi_j - \phi_j/\gamma_j$. The bound $|q \operatorname{sech}^2 q - \tanh q| \leq \frac{2}{3}|q|^3$ gives

$$\partial_{\gamma_j} L = \frac{c_j}{\gamma_j} \partial_{c_j} L + c_j \langle r, \zeta_j \rangle, \quad \|\zeta_j\| \leq \frac{2}{3} \gamma_j^2 M_{3,j}. \quad (7)$$

Readout fitting therefore removes the leading amplitude-like scale force. The remainder is controlled for small preactivations, but its useful sign and persistence remain to be established [4, Section 6.2].

4 Why bounded scales can make further precision expensive

For the matrix A on $x_i \in [-1, 1]$, suppose every real slope satisfies $|a_j| \leq \Gamma$; real biases are unrestricted. Here Γ is a cap, not a predicted plateau. Define

$$\rho_\Gamma = \frac{\pi}{4\Gamma} + \sqrt{1 + \frac{\pi^2}{16\Gamma^2}}, \quad E_k(\Gamma) = \frac{2\rho_\Gamma^{-k}}{\rho_\Gamma - 1}.$$

Every feature is analytic and bounded by one on the same Bernstein ellipse. Degree- k Chebyshev truncations therefore give a rank- $(k+1)$ surrogate with error at most $\sqrt{W}E_k(\Gamma)$ [4, Sections 1–2]. Hence, when the indices exist,

$$\sigma_{k+2}(A) \leq \sqrt{W}E_k(\Gamma), \quad \kappa_2(A) \geq \frac{\rho_\Gamma - 1}{2\sqrt{W}} \rho_\Gamma^{W-1}. \quad (8)$$

The condition-number bound assumes $m \geq W+1$ and full column rank. Since $\log \rho_\Gamma \sim \pi/(4\Gamma)$ for large Γ , it exposes an exponential factor in W/Γ . This is a classical analytic-approximation argument specialized to $\tanh$.

Conditioning matters only if the target needs the weak directions. Let $\delta_k(y)$ be its best normalized sample error in degree- k polynomials. Uniformly over all dictionaries under this cap,

$$\boxed{R \geq [\delta_k(y) - E_k(\Gamma) \|c\|_1]_+}. \quad (9)$$

Thus extra accuracy may require a large coefficient budget. This is approximate/numerical rank loss, not necessarily exact rank loss or an unavoidable function-error floor [4, Sections 3–4].

5 How much scale drift remains after the residual falls?

For ordinary raw-parameter gradient flow in (1), $|x_i| \leq X$ and Cauchy–Schwarz give $|\partial_{a_j} L| \leq X|c_j|R$. Consequently, even with joint center and readout motion,

$$|\gamma_j(T) - \gamma_j(0)| \leq X \int_0^T |c_j(t)| R(t) dt. \quad (10)$$

This uses unit-rate geometry flow; GD replaces the integral by $\sum_n \ell_n |c_{j,n}| R_n$. If $|c_j| \leq C_{\text{out}}$ and $R(t) \leq R(0)$, reaching $\Gamma > \gamma_j(0)$ requires at least $(\Gamma - \gamma_j(0))/(XC_{\text{out}}R(0))$ time. For $\Gamma = \lambda_*/h$, fixed initial scales, and bounded constants, this is order h^{-1} , or h^{-2} when frozen readouts are $O(h)$.

A more direct plateau interpretation follows if $R(t) \leq R_0 e^{-\mu t} + R_{\text{floor}}$:

$$|\gamma_j(T) - \gamma_j(0)| \leq XC_{\text{out}} \left[\frac{R_0}{\mu} (1 - e^{-\mu T}) + TR_{\text{floor}} \right]. \quad (11)$$

An integrable residual gives finite total slope travel. A small nonzero floor instead permits a long finite-budget plateau. These bounds retain the readout magnitude; coefficient growth can change the conclusion.

When an exponential bound applies. For an independently optimized centered scale, suppose the whole-line residual assumptions of (3) persist until the first hit of Γ , with $0 < \gamma_0 < \Gamma \leq \Omega$, $|c| \leq C_{\text{out}}$, and $M_\Omega(t) \leq M$. Integrating the reciprocal speed bound yields

$$T_\Gamma \geq \frac{1}{C_F C_{\text{out}} M \Omega} \int_{\gamma_0}^{\Gamma} s^3 e^{\pi \Omega / (2s)} ds. \quad (12)$$

If $\Gamma \geq 2\gamma_0$, this is at least $\gamma_0^4 e^{\pi \Omega / (4\gamma_0)} / (C_F C_{\text{out}} M \Omega)$. This is exponential in Ω/γ_0 , not automatically in width. Proving spectral persistence—despite low-frequency leakage, boundaries, and joint updates—is the substantive missing step.

6 Conclusion and status

The proposed mechanism is a **mismatch between residual refinement and scale acquisition**. It becomes a precision barrier only when the surviving error needs the weak directions of the bounded-scale dictionary. The next theorem must connect the controlled coarse-fitting phase to persistent weak scale sensitivity and a target-dependent accuracy bound.

Result	Status
Fourier and projection identities	Established tools; frequency and domain/rank assumptions matter.
Toy allocation; bounded-slope spectrum	Exact model and explicit specializations of classical arguments.
Scale-drift / exponential bounds	Proved conditional on the stated trajectory and coordinate assumptions.
The full precision-plateau mechanism	Not yet proved for joint tanh GD; no universal impossibility claim.

References

- [1] Z.-Q. J. Xu et al. *Frequency Principle: Fourier Analysis Sheds Light on Deep Neural Networks*. Commun. Comput. Phys. 28 (2020), Section 6. [arXiv:1901.06523](#).
- [2] Y. Wang and C.-Y. Lai. *Multi-stage Neural Networks: Function Approximator of Machine Precision* (2023), Section 2.2. [arXiv:2307.08934](#).
- [3] D. P. O’Leary and B. W. Rust. *Variable Projection for Nonlinear Least Squares Problems*. Comput. Optim. Appl. 54 (2013), 579–593. [doi:10.1007/s10589-012-9492-9](#).
- [4] *Bounded slopes, spectral saturation, and readout scale*. Companion proof note, Sections 1–4 and 6.2 (supplied separately).